

Econ 2 - Lecture 4 - 1/14/26

Lecture Quiz #2 Released Today, Due Wednesday, 1/21

No Class Monday, MLK Day on 1/19

Coffee Hour #1 Tomorrow, 2-3 PM

Discussion Activity #1 This week, Activity #2 next week

Midterm Exam on Monday, February 2nd, Chapters 1, 2, 3

Today: GDP (Chapter 3.1), Next Class: GDP & CPI (3.2)

Last Class Recap

GDP: The market value of all final goods and services produced for a market place during a given period of time, within a country's borders

Market Value: convert production into a dollar amount

Final Goods/Services: Do NOT count intermediate goods

Intermediate Goods: used to produce final goods (Best Buy computer)



Produced: goods/services to be created/new

→ stocks, bonds, land → not creations, not counted

For a market place: consumers/firms gathering to buy/sell products → tracked in a formal setting

→ Informal market place: ~ 2 Trillion / year

→ Babysitting, lawn care, volunteering → not counted

Given Period of Time: Isolate the date of Production

Car produced and sold in 2020

↳ Entire market price of car counted in 2020

In 2025 → Buy 2020 used car, already counted in 2020!

↓
Value of car not counted in 2025

What is "new" in 2025?

Service of buying at a dealership → "safety"

"Created" in 2025 → Part of 2025 GDP

Existing homes → not counted in GDP → Real Estate fees
one

Within a country's borders

Domestic → Border → GDP

Gross National Product (GNP)

Production based on nationality

Sarah → Lives in UK → Part of UK GDP

US Citizen → Part of USA GNP

Measure GDP? Bureau of Economic Analysis

Categorize GDP: Expenditure Approach

1.) Consumption (C): household

2.) Investment (I): firms

3.) Government Purchase (G): government

4.) Net Exports (NX): foreign entities

$$GDP = Y = C + I + G + NX$$

Consumption (C) = 67% of GDP

Goods: item we gain ownership of (tangible)

Durable: long-lasting goods (at least 3yrs)

Non-Durable: frequently bought

→ food, gas, clothes, etc.

Services: Intangible items that we
(~45%) do not gain ownership of
of GDP

→ Subscription

→ education

→ Rental Housing

→ Airline

→ Concerts / entertainment

→ Restaurants → service of a meal

→ Health care

Housing in GDP $\rightarrow$ 12%

Rent = Service $\rightarrow$ into C component of GDP

Owning $\neq$ Service $\neq$ C of GDP?

Create imputed rent:

Estimate the value of every occupied dwelling on a monthly basis $\rightarrow$ add to GDP

10 homes on a street

Each home could rent for \$1,000/month

Add 10 homes $\times$ \$1,000 to GDP each month

New homes are not Consumption $\rightarrow$ Investment!

Investment (I): Goods/services bought by firms
(~18% of GDP)

1. Non-Residential Investment $\rightarrow$ Capital

$\rightarrow$ Tractor, machine, office desks, software,
plant / building, etc.

2. Residential Investment: New Home built

$\rightarrow$ Durable, fixed (non-moving) asset

$\rightarrow$ Produce "Home Goods"

3. Change in Private Inventory (~1-0%)

Example: Inventory = unsold goods

Campus Bookstore in 2024: \$10,000 in Hoodies

2024: GDP increased by \$10,000

Imagine only \$8000 in hoodies are bought in 2024

Add \$8000 to Consumption in 2024

Remaining \$2000 → Unsold, added to inventory
→ Added to Investment in 2024

In 2025 → sell the "leftover" \$2000 in hoodies

Add \$2000 in Cons. in 2025

Remove \$2000 in Hoodies from inventory

Inventory decreases by \$2000, Investment ↓ \$2000

Overall GDP Change in 2025 = \$0

Gov't Purchases (G)
(~17%)

Purchases by government →

Necessities that are to profit from

→ Protection: National Defense, Police, Fire

→ Infrastructure, education

Note: Transfer Payments = Welfare, Social Security, etc.
Not Counted → no production

Net Exports (NX) = Exports (X) - Imports (M)
→ Money is coming in → Money is leaving
→ Goods leaving → Goods entering

$$\begin{array}{ccccccc}
 Y & = & C & + & I & + & G & + & NX \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & & \begin{array}{l} \swarrow^x \searrow^m \\ 3.36T \quad 4.12T \end{array} \\
 & & 21.11T & & 5.42T & & 5.32T & & \\
 & & & & & & & & \underbrace{\hspace{1cm}} \\
 31.095T & & & & & & & & - 0.76T
 \end{array}$$

$\downarrow$
 Production
 Firm Profit
 Household Income

Calculate GDP

Product	2009		2016	
	<u>Q</u>	<u>P</u>	<u>Q</u>	<u>P</u>
Food	1000	1	1500	2
Housing	100	100	150	150
Movies	500	5	1000	10

$$GDP_{2009} = Q_{09}^F \times P_{09}^F + Q_{09}^H \times P_{09}^H + Q_{09}^M \times P_{09}^M$$

$$\begin{array}{l}
 \text{Nominal} \\
 \underline{GDP_{2009}}
 \end{array}
 = 1000 \times 1 + 100 \times 100 + 500 \times 5 = \$13,500$$

→ amount of dollars transacted in 2009

$$\begin{array}{l}
 \text{Nom.} \\
 \underline{GDP_{2009}}
 \end{array}
 = Q_{09} \times P_{09}$$